

Part I

FLOW MATCHING

TABLE OF CONTENTS

- 1 Motivation and Background 4
- 2 Flow Matching Principle 5
- 3 Conditional Flow Matching Objective 6
- 4 Training and Sampling Algorithms 7

MOTIVATION AND BACKGROUND

Continuous flows

- ▶ A continuous flow transforms samples by solving an *ODE*:

$$\frac{dX_t}{dt} = v_t(X_t), \quad X_0 \sim p_0.$$

- ▶ The vector field v_t defines how each particle moves from a simple base distribution p_0 toward the data distribution p_1 .
- ▶ Generation becomes deterministic: sample $X_0 \sim p_0$, integrate the ODE, and output X_1 .

Origin and intuition

- ▶ It reuses *ODE-based generation* from CNFs (Chen et al., 2018), *noise-to-data paths* from diffusion models Song et al. (2021), and *mass-transport geometry* from optimal transport.
- ▶ Main motivation: train continuous-time generative models *without simulating the ODE during training*, unlike likelihood-based CNF training.
- ▶ It learns, by *regression*, a time-dependent *velocity field* $v_\theta(t, x)$ transporting Gaussian noise to the data distribution.
- ▶ It was formalized through *Flow Matching* Lipman et al. (2023), *Rectified Flow* Liu et al. (2022), and *Stochastic Interpolants* Albergo and Vanden-Eijnden (2023).

MOTIVATION AND BACKGROUND

Continuous flows

- ▶ A continuous flow transforms samples by solving an *ODE*:

$$\frac{dX_t}{dt} = v_t(X_t), \quad X_0 \sim p_0.$$

- ▶ The vector field v_t defines how each particle moves from a simple base distribution p_0 toward the data distribution p_1 .
- ▶ Generation becomes deterministic: sample $X_0 \sim p_0$, integrate the ODE, and output X_1 .

Transport viewpoint

- ▶ The ODE induces an evolution of the density p_t , since the dynamics are deterministic, probability mass is only *transported*: it is neither created nor destroyed. This gives the *continuity equation*:

$$\partial_t p_t(x) + \nabla \cdot (p_t(x) v_t(x)) = 0.$$

- ▶ This is the deterministic analogue of the *Fokker-Planck equation*. For an SDE, an additional diffusion term appears; for an ODE, only the transport term remains.
- ▶ Flow matching learns the velocity field v_t whose induced density evolution moves noise toward data.

FLOW MATCHING PRINCIPLE

Ill-posed target

Many velocity fields can transport p_0 to p_1 . From samples alone, the problem is *ill-posed*: we only observe the endpoints, not the intermediate path p_t nor the velocity $v_t(x)$. Flow matching makes the problem tractable by choosing simple *conditional paths* whose velocities are known.

Choose a path, not directly a vector field

- Choose an interpolation, together with the coupling of (X_0, X_1) .

$$X_t = \psi_t(X_0, X_1), \quad X_0 \sim p_0, \quad X_1 \sim p_1 \quad (\text{e.g. } X_t = (1-t)X_0 + tX_1).$$

- This choice defines both the intermediate distributions p_t and the conditional velocity

$$U_t = \partial_t \psi_t(X_0, X_1) \quad (\text{e.g. } U_t = X_1 - X_0).$$

The conditional trick

- For each hidden pair $Z = (X_0, X_1)$, the velocity U_t is known.
- The network only sees (t, X_t) , not Z .
- Therefore, regression learns the average velocity of all paths crossing x : $v_t(x) = \mathbb{E}[U_t \mid X_t = x]$.

CONDITIONAL FLOW MATCHING OBJECTIVE

Marginal loss: ideal but inaccessible

- Ideally, one would regress on the true marginal velocity field:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, X_t} \left[\|v_\theta(t, X_t) - v_t(X_t)\|^2 \right].$$

- But $v_t(x)$ is unknown, so this loss cannot be used directly.

Conditional loss: tractable with the same gradient

- Use the known conditional velocity instead:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, X_0, X_t} \left[\|v_\theta(t, X_t) - U_t\|^2 \right].$$

- The key identity is

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathcal{L}_{\text{FM}}(\theta) + C, \quad C \text{ independent of } \theta, \quad \text{therefore} \quad \nabla_\theta \mathcal{L}_{\text{CFM}}(\theta) = \nabla_\theta \mathcal{L}_{\text{FM}}(\theta).$$

- Indeed since $v_t(X_t) = \mathbb{E}[U_t | X_t]$, the residual $U_t - v_t(X_t)$ is orthogonal to any function of (t, X_t) , in particular to $v_\theta(t, X_t)$. Hence the cross term vanishes.

TRAINING AND SAMPLING ALGORITHMS

Algorithmic view

- During training, the ODE is *not* solved: we only regress on known velocities along sampled conditional paths.

Algorithm 1 Conditional flow matching training

- 1: sample $X_1 \sim p_1, X_0 \sim p_0, t \sim \mathcal{U}(0, 1)$
 - 2: set $X_t = (1 - t)X_0 + tX_1$
 - 3: set $U_t = X_1 - X_0$
 - 4: minimize $\|v_\theta(t, X_t) - U_t\|^2$
-

- During sampling, the learned field is integrated as an ODE to transport noise into data.

Algorithm 2 Sampling with the learned flow

- 1: sample $X_0 \sim p_0$
 - 2: solve $\frac{dX_t}{dt} = v_\theta(t, X_t)$ from $t = 0$ to $t = 1$
 - 3: **return** generated sample X_1
-

REFERENCES FOR THIS PART I

-  Albergo, M. S., & Vanden-Eijnden, E. (2023). **Building normalizing flows with stochastic interpolants.** *International Conference on Learning Representations.*
-  Chen, R. T. Q., Rubanova, Y., Bettencourt, J., & Duvenaud, D. K. (2018). **Neural ordinary differential equations.** *Advances in Neural Information Processing Systems.*
-  Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., & Le, M. (2023). **Flow matching for generative modeling.** *International Conference on Learning Representations.*
-  Liu, X., Gong, C., & Liu, Q. (2022). **Flow straight and fast: Learning to generate and transfer data with rectified flow.** *arXiv preprint arXiv:2209.03003.*
-  Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2021). **Score-based generative modeling through stochastic differential equations.** *International Conference on Learning Representations.*